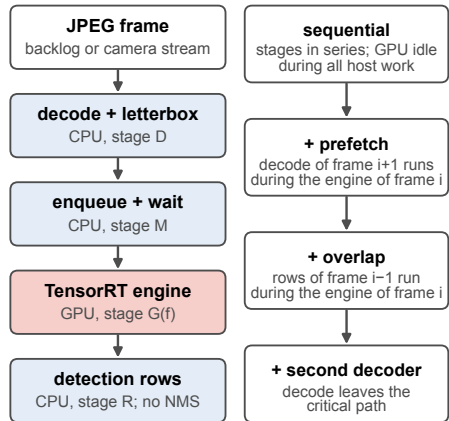
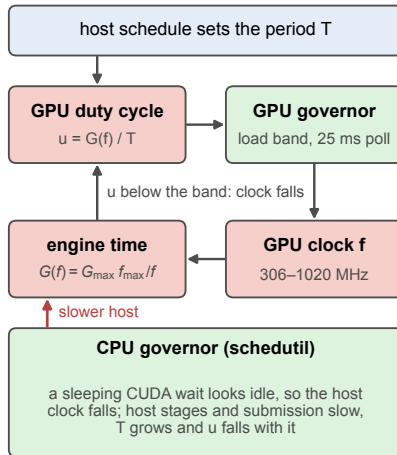


(a) Detection pipeline and schedules



(b) Two governors close the loop



(c) Measurement and replication

Power

INA3221 VDD_IN at 20 Hz; energy per image

Clocks and load

GPU devfreq, CPU cpufreq, driver GPU load

Accuracy

COCOeval on 5,000 val2017 images; sha256 of every prediction file

Latency

frame release to finished rows; p50, p95

Jetson Orin Nano Super

GPU, TensorRT; main study

Raspberry Pi 5 + Hailo-8

NPU without user clock control

Rubik Pi 3 (QCM6490)

CPU and Hexagon NPU